G H International Skill Tech University, Pune.

Department of Examinations

Question Paper

Continuous Assessment Examination-II (CAE-II)-Winter 2024

Programme: B Tech **Year: First** **Semester: I**
Max Marks: 20 **Duration: 60 mins.**
Course Name: Basic Electrical Engineering **Course Code: MT103**

Instructions:

1. You are permitted to use calculators during the exam.
2. Clearly label each question and ensure your answers are organized.
3. Utilize diagrams and equations where relevant, and always include units
4. Students must label their answers with the corresponding question number,such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	To develop students' ability to solve basic electrical engineering problems through analytical thinking and application of theoretical concepts.
CO2	Interpret and analyze data from practical experiments and simulations to draw conclusions about electrical phenomena on transformer.
CO3	Demonstrate awareness of electrical safety practices and regulations in the context of designing, testing, and operating electrical circuits and systems.

Q. No.	Sub Q. No.	Question	Marks
1		Solve any **TWO**	
	A	I.Explain the working principle of a transformer. II.How do core and shell-type transformers differ in terms of construction and performance?	05
	B	Derive the EMF equation of primary and secondary of a transformer. What are the factors that affect the induced EMF in the transformer windings?	05
	C	I. Discuss the differences between an ideal and a practical transformer with example. II. Explain losses in a transformer in details.	05
	D	I.Define the term MMF, Reluctance, Permaence and Suceptance in details II Np= 100 & Ns= 200 , Ep= 220 , Ip= 10 A Find Es and Is , K and type of transformer	05
2		Solve any TWO	
	A	Explain the working principle of a semiconductor PN junction diode.	05
	B	What is Sinusoidal Modulation? Explain it with an example.	05
	C	Describe the forward and reverse bias characteristics of a PN junction diode.	05
	D	Describe the working principle of a Silicon Controlled Rectifier (SCR).	05